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Integrated scheduling of production and material delivery for the intelligent manufacturing system

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ABSTRACT

This paper addresses the issue of integrated scheduling of production and material delivery in an intelligent manufacturing system. The system includes multiple flexible flowlines and a fleet of automated guided vehicles (AGVs), whose capacity is shared among the flowlines. Multiple orders from different customers are processed on the flowlines, with AGVs delivering materials in different batch sizes based on varying product quantities of orders and limited load capacity. The problem is formulated as a mixed-integer linear programming model to minimise total order delays and the number of used AGVs. To handle the complexity caused by the uneven temporal coupling relationship between production and material delivery, a bi-level neighbourhood search solution algorithm is proposed. This algorithm uses variable neighbourhood search for production scheduling and adaptive large neighbourhood search for material delivery scheduling. Extensive numerical experiments using real-case-based instances are conducted to demonstrate the satisfactory performance of our proposed algorithm. Furthermore, we emphasise the advantages of integrated scheduling compared to traditional sequential scheduling across various problem settings.

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KEYWORDS

Flexible flowlines; limited delivery capacity; integrated scheduling; bi-level algorithm; neighbourhood search

1. Introduction

In today's era of Industry 4.0, advanced technologies such as cyber-physical systems (CPS), Internet-of-Things (IoT), and cloud computing have the potential to revolutionise manufacturing (Guo et al. 2020). One example is the widespread adoption of automated and flexible equipment to tackle complex tasks (Parente et al. 2020). For instance, due to their high speed, accuracy, and low cost, automated guided vehicles (AGVs) are widely used to prevent human error and contamination in industries such as semiconductors, cosmetics, household appliance (Elmi and Topaloglu 2014). The increasing controllability of intelligent production and logistics equipment provides a reliable foundation for leaner schedules, boosting productivity and reducing operational costs. In addition, the new connectivity among smart equipment and modularity of production lines such as the widespread application of reconfigurable flowlines, enable mass customisation (Dou et al. 2021; Kim and Lee 2023). Consequently, the variety of product series and specifications has proliferated in recent years, allowing manufacturers

to enhance customer satisfaction by offering customised products and services (Singh and Sarin 2024). Moreover, to effectively adapt to market fluctuations and minimise inventory costs, many businesses have shifted from traditional make-to-stock mass production to the make-to-order (MTO) production mode (Zennaro et al. 2019).

Although the advantage of adopting these frontier technologies is obvious, the actual effects rely heavily on the corresponding production management (Li and Huang 2021). Specifically, scheduling is a key management focus for both industry and academia, and it becomes more challenging in the context of intelligent manufacturing compared to traditional scenarios (Parente et al. 2020). Firstly, the complexity is increased by the growing product variety and parallel production across multiple production lines. Secondly, the coordination between various constituent sectors within intelligent manufacturing systems must be considered. Additionally, the large problem scale presents a common difficulty, as intelligent manufacturing factories tend to be large-scale, featuring multiple flowlines to leverage economies

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of scale. To address these issues and realise the potential improvements, efficient scheduling approaches must be developed (Frazzon et al. 2018).

One common problem faced by the manufacturing factories is the order delay caused by the mismatch between the production and material delivery pace, since the production planning and material delivery control are typically two sequential phases of production organisation. For example, a sequential scheduling approach is applied in the 6-flowline-intelligent factory of our collaborating enterprise – a well-known air conditioning compressor manufacturer. After receiving orders, the production planning department makes the production schedule first; and then, material delivery tasks are organised by an AGV scheduling system based on simplistic heuristic rules during the production process. Despite the advanced automated equipment in its factory, interruptions are always caused by the sequential scheduling. One possible reason is that material demands fluctuate widely when producing various production specifications, with the fact that the types and specifications proliferate in the household appliance industry due to highly customised demands. In addition, the type and quantity of materials used for producing different products are different. Therefore, frequent production switch may result in imbalance in the material delivery requirement. When the material demand fluctuates greatly, existing AGV capacity may be inadequate at times of high demand, causing delays in material delivery and production, while idle at times of low demand, causing extra operational costs. The parallel production on the 6 flowlines is another potential reason for current production disorganisation. When the material demand on several flowlines centralised at certain time slots, the imbalance of material delivery is amplified comparing to single flowline system. In production systems where parallel production of various production specifications takes place and AGVs are shared across flowlines, integrated production and material delivery scheduling becomes quite necessary.

The interplay between production and in-plant material delivery can be summarised bidirectionally. On one hand, on-time material delivery is a prerequisite for executing a production schedule. Without proper material delivery scheduling, materials may arrive at workstations later than the scheduled production time, causing interruptions. On the other hand, production scheduling influences the complexity of material delivery scheduling. Since the expected material arrival time is determined by the production schedule, the material delivery scheduling will become quite difficult if the delivery requirements are centralised in the time dimension. This motivates our study on the integrated scheduling methods for these two processes.

Particularly, this paper focuses on the intelligent manufacturing of production and material delivery for the intelligent flowline system. Motivated by our collaborating air conditioning manufacturer, we consider a system consisting of heterogeneous flowlines with different capable production types and cycle times. Each flowline comprises multiple workstations where various operations take place. At each workstation, there are demands for various materials that may have to be satisfied by multiple batch deliveries. The delivery batches are divided based on the quantities of products and the load capacity of the AGVs.

Apart from the household appliance manufacturing industry, the abovementioned problem also exists in many other industries. Intelligent assembly systems, such as those in electronic product manufacturing factories, are typical examples. In these systems, a variety of products are produced on multiple parallel flowlines at a high pace. Additionally, different types of materials serving different processes are delivered by AGVs from warehouses to workstations in batches. If the required materials arrive unpunctually, the production process of the flowlines will be interrupted, leading to higher operation costs and lower production efficiency. Due to the interdependence of production and material delivery in such systems, this impact cannot be ignored. Managers should consider an integrated scheduling method to coordinate these two subproblems effectively. Furthermore, for industries that emphasise the clean production environment, such as semiconductors, cosmetics, and pharmaceuticals, integrating the scheduling of production and material delivery is essential to fully utilise the configured automated equipment, i.e. AGVs.

The superiority of integrated scheduling of production-logistics processes has been demonstrated in the literature (Battini et al. 2017; Chen and Vairaktarakis 2005; Scholz-Reiter, Frazzon, and Makuschewitz 2010). Nevertheless, our research problem is distinct and challenging in two aspects. First, the asymmetric temporal correlation between flowline production tasks and the material delivery tasks complicates problem modelling. Specifically, unlike the one-to-one relationship, i.e. a production process uniquely corresponds to a transportation process (either happened outside the plant or between two in-plant workstations) considered in the main streams of related literature, the production-logistics correlation has to be modelled by groups. Furthermore, the asymmetric temporal constraints expand the feasible region, making the integrated scheduling of production and material delivery problem (ISPM DP) more difficult to solve. Second, incorporating realistic factors including mixed flow production, parallel flowline, and material delivery in batch, increases the problem scale and amplifies the difficulty in problem-solving. For instance, the

number of batches required at each workstation can vary for a production specification. Simultaneously, production processes and material requirements differ across product specifications. The heterogeneous density of the material delivery requirements for different products necessitates integrating production and material delivery scheduling to smooth the requirements. As each production process may require batches of material deliveries, the problem scale escalates rapidly with increasing orders, product specifications, and number of production lines.

The contributions of this paper are as follows. First, we consider a practical scenario within the context of intelligent manufacturing, involving multi-type products and multi-order processing across multiple flowlines. We formulate a mixed integer linear programming (MILP) model to address the ISPM DP, a topic that receives scarce attention in the related literature. Second, as both subproblems, i.e. MFFL production scheduling problem and in-plant material delivery scheduling problem, are known to be NP-hard (Reza Hejazi and Saghafian 2005; Savelsbergh 1985), we design a bi-level heuristic algorithm based on neighbourhood search to address the complexity and scalability issues associated with large-scale instances. At the upper level, the assignment of production orders to each flowline and the determination of production sequences are managed through a variable neighbourhood search (VNS) algorithm, which includes a biased-random-key-crossover (BRKC) operator and a VND scheme. At the lower level, we handle the assignment of material delivery tasks to AGVs and the optimisation of delivery sequences using an adaptive large neighbourhood search (ALNS) algorithm with several tailored operators. Third, to evaluate the effectiveness of our proposed solution approach, we conduct extensive numerical experiments on the real data collected from our collaborating air conditioning compressor manufacturer in Nanchang, China. The results demonstrate the superiority of integrated scheduling over sequential production and material delivery scheduling, highlighting the necessity for managers to make integrated schedules to leverage the potential production efficiency of intelligent manufacturing. According to the sensitivity analyses, it is evident that when AGVs are relatively inadequate, i.e. small configured quantity or long delivery time per task, managers are recommended to make integrated schedules to smooth the material requirements across the time dimension to avoid production interruption as much as possible. In addition, our results show that integrated scheduling is more cost-effective in reducing order delays than simply increasing investments in hardware equipment. Furthermore, our bi-level solution algorithm, which generates high-quality feasible solutions within

acceptable time and outperforms Gurobi in most tested cases, offers an easy-to-use way for applying integrated scheduling in practical scenarios.

The remainder of this paper is organised as follows. Section 2 conducts a comprehensive review of relevant literature. Section 3 describes the problem and presents the MILP model formulation. Section 4 proposes a bi-level optimisation algorithm. In Section 5, we carry out extensive numerical experiments to assess the effects of various parameters and to show the performance of our proposed algorithm. The conclusions of this paper and discussions of future directions are in Section 6.

2. Literature review

Flowline production remains a prevalent mode due to its evolving automation and flexibility, and its scheduling has received extensive attention in prior research (Komaki, Sheikh, and Malakooti 2019; Naderi, Gohari, and Yazdani 2014; Perez-Gonzalez and Framinan 2024). Meanwhile, production logistics plays a pivotal role in the production system, significantly impacting overall production efficiency (Nyhuis and Wiendahl 2009). Literature concerning production logistics scheduling is typically categorised into four main areas: scheduling for the procurement logistics of raw materials, in-plant transportation of work-in-process (WIP) products, material delivery, and the delivery of finished goods to customers (Arviv, Stern, and Edan 2016; Fathi et al. 2016; Pelletier, Jabali, and Laporte 2016; Ramanathan, Subramanian, and Cheikhrouhou 2015). Our work falls into the third category. With the escalating market demand fluctuations, the influence of production logistics on production efficiency has gained prominence, necessitating research on ISPM DP (Lee and Chen 2001). This section reviews the literature focusing on three primary types of integrated production-logistics scheduling studies.

2.1. Integrated scheduling of production and inbound/outbound logistics

Inbound and outbound logistics, encompassing aspects like raw material procurement and finished goods distribution, exert substantial influence on production efficiency driven by factors such as just-in-time production management and product perishability, etc. (Chen 2010). In recent years, a multitude of researchers has delved into the area of integrated scheduling of production and inbound/outbound logistics. Lee and Chen (2001) prove the NP-hardness of integrated scheduling of the production on a two-stage flowline and product delivery by a single-capacity vehicle. Tang and Gong (2009) tackle a similar problem and investigate a production system

featuring a batch processor and a single-capacity vehicle. They introduce a pseudo-polynomial time algorithm to address the coordinated scheduling challenge of production and transportation to minimise total completion time and processing costs. Scholz-Reiter, Frazzon, and Makuschewitz (2010) formulate the inter-organisational integrated production and routing problem as a mixed integer programming (MIP) and demonstrate the benefits of integrated scheduling on cost and lead time reduction through their computational analysis. Meanwhile, due to the intractability of the MIP model, they propose that powerful heuristic algorithms are needed for solving larger instances for further exploration of the effects of integrated scheduling. Chevroton et al. (2021) design an integrated solution approaches based on heuristics and branch-and-bound method for production scheduling and finished goods delivery routing, which outperforms the greedy sequential algorithm in reducing inventory and delivery costs. Considering a flow shop scenario where transportation of intermediate products is critical for the consecutive production steps at different places along the supply chain, the superiority of integrated scheduling compared with the sequential scheduling in respect of more performance indicators (storage, unpunctuality, lead-time and total cost) is elaborated by Ehm and Freitag (2016).

Some researchers work on more complex problems. For instance, Geismar, Dawande, and Sriskandarajah (2011) explore optimisation strategies for the collaboration between the production and the transportation of finished perishable products under diverse settings, including transporter numbers, production rates, and various objectives such as make span and total completion time. Ullrich (2013) extends the scope by considering parallel manufacturing machines and heterogeneous transporters, developing MILP models for both production scheduling and material delivery scheduling. Recognising the NP-hardness of both subproblems, they propose a genetic algorithm (GA) to solve the integrated problem. Wang and Cheng (2009) embrace a holistic approach, addressing the collaboration of both inbound and outbound logistics alongside the production process. They optimise batch patterns for a single capacitated vehicle during each logistics phase to minimise the arrival time of the last delivered job to the customer.

Among existing literature, the unidirectional interaction between production and inbound/outbound logistics (or vehicle routing) is commonly described by constraints like 'production of an order cannot commence until the required raw materials arrive' or 'the delivery of orders cannot initiate before the production completion time'. However, the interplay between production and in-plant logistics is bidirectional, i.e. (1) on-time material

delivery is an essential prerequisite for the execution of a production scheduling; (2) production scheduling influences the complexity of material delivery scheduling. Consequently, the temporal coupling relationship is more intricate and a new model has to be proposed to solve the ISPM DP.

2.2. Integrated scheduling of production and WIP transportation

Several studies have concentrated on the integration of production and WIP transportation scheduling within job shops. Bilge and Ulusoy (1995) develop an MIP model aimed at creating a unified schedule for job shop production and the delivery of WIP products. This model considers several single-capacity AGVs and specific processing procedures. To address a similar problem, Lacomme, Larabi, and Tchernev (2013) introduce an extended disjunctive graph, where the transportation of WIPs between workstations is treated as an integral part of the production process. They propose a memetic algorithm with a solution evaluation procedure based on Bellman's longest path algorithm to tackle this challenge. Zheng, Xiao, and Seo (2014) concentrate on minimising make span and design a heuristic algorithm based on tabu search (TS) for simultaneous machine and AGV scheduling. Their numerical results indicate a 17% improvement on make span in comparison with the traditional sequential scheduling method. Significant improvements of integrated scheduling are also reported by Berterottière, Dauzère-Pérès, and Yugma (2024).

In addition to traditional job shop production characterised by specific processing procedures, some studies explore more flexible production environments featuring multifunctional machines. The heuristic solution algorithms are extensively explored in this setting due to the problem complexity. For example, Nouri, Driss, and Ghédira (2016) delve into the collaboration of production and WIP product transportation in a job shop with flexible machines and capacitated vehicles. They employ a clustered holonic multiagent model, incorporating GA and TS for global and local search, respectively. Homayouni and Fontes (2021) devise an effective local-search-based heuristic to deal with the production and transportation scheduling in a flexible job shop system. Pan et al. (2023) take advantage of learning algorithms to improve the performance of traditional evolutionary optimisation and achieve the exploration-exploitation balance by the competition and cooperation of the subpopulations: elite subpopulation, exploitative subpopulation, and exploratory subpopulation.

It is worth noting that due to the discrete nature of job shop production, the literature mentioned above often

assumes that each delivery trip carries a single workpiece. Consequently, there is typically a symmetric relationship between a production task and a delivery task from the preceding machine to the current one and from the current machine to its succeeding one. In this case, the constraints on both production and delivery tasks are of similar structure, i.e. for each sequence, the succeeding task cannot be implemented until its preceding task finished, and AGVs can even be regarded as a production machine when the no-load time is ignored. However, the ISPM DP is more challenging to solve. First, in mixed flow flowline production systems, the interaction between the production tasks and in-plant material delivery tasks is asymmetric. For a production task, the accomplishment of the corresponding material delivery task is necessary for its processing, while the implementation of a material delivery task is independent from the production process. Therefore, the temporal constraints between the flowline production and material delivery, as compared to those between WIP transportation, is much looser, leading to a larger feasible region. Moreover, in terms of the problem scale, since an order always requires several batches of material delivery, which differ over product types and production processes, the ISPM DP tends to be more complicated than the integrated scheduling problems of production and WIP transportation. For above reasons, the models and algorithms devised to solve integrated job shop scheduling problems cannot be directly applied to achieve integration in our concerned scenarios.

2.3. Integrated scheduling of flowline production and material delivery

In contemporary flowline systems, the transportation of workpieces between sequential workstations is primarily facilitated by conveyor belts, while the delivery of raw materials from warehouses to workstations is managed by vehicles. Given the need to meet the material requirements of various workstations with limited transportation resources, researchers have explored the integrated scheduling of flowline production and in-plant material delivery.

One particular type of flowline, the assembly line, holds a prominent position in industrial operations and has been the subject of extensive research (Battini et al. 2017). Sternatz (2015) establishes a practice-oriented model considering limited line-side buffer and employs the enhanced multi-Hoffmann heuristic to address the assembly line balancing and material delivery scheduling problem for minimising the total duration of production and material delivery. Battini et al. (2017) consider a two-stage production process for homogeneous workpieces

and investigate two prevalent alternative methods for parts supply: direct supply from the warehouse or indirect supply through an intermediate supermarket. Their detailed case study on the assembly of a self-priming pump highlights the necessity of simultaneous decisions on assembly line balancing, supply mode of parts and scheduling of the supermarket area. Subsequently, Rahman, Janardhanan, and Nielsen (2020) deal with the challenges of assembly line balancing and material delivery scheduling through a hierarchical heuristic-based integrated decision approach. This approach proves beneficial in terms of minimising cycle time and total delays. Yilmaz (2022) introduces an ε -constrained approach to address the integrated scheduling problem, focusing on minimising total operation costs and achieving workload balance in a U-shaped assembly line. Chen, Jia, and He (2023) propose a bi-level multi-objective GA to reduce the workstation count and enhance workload smoothness in an assembly line production system. Even though the literature above concerns the material delivery scheduling together with the assembly line balancing, the sequencing of production orders and the asymmetric interaction between production and material delivery garners relatively less attention.

As discussed above, the integration of production and in-plant material delivery is a vital and efficient response to the increasingly challenging manufacturing systems. However, to the best of our knowledge, there is a lack of literature on integration within flowline systems where continuous production heavily relies on the punctual delivery of materials. Our research differs from previous work in the following aspects. First, the material requirement of an order at each workstation is fulfilled by several delivery tasks. Second, the temporal coupling relationship between the production task and its corresponding material delivery task is asymmetric. Third, mixed flow production and parallel flowline production are considered simultaneously.

3. Problem description and mathematical modelling

3.1. Problem description

We consider a large-scale intelligent factory with heterogeneous flexible flowlines $\mathbb{L} = \{1, \dots, L\}$ (indexed by l) and H homogenous AGVs. Each flowline is equipped with balanced workstations $\mathbb{I}^l = \{1, \dots, I^l\}$ (indexed by i), each of which is responsible for several processes. The factory applies MTO production mode and receives a set of customer orders $\mathbb{J} = \{1, \dots, J\}$ (indexed by j) with due dates $\{d_j, \forall j\}$ during the scheduling horizon. We assume each order contains only one type of product (If an order

contains more than one type of product, it can be divided into several orders with one type of product in each). Between the production of order j and j' on flowline l , switch time $e_{jj'}^l$ is required for changing moulds and set-up. Each flowline is capable of producing all types of the company's products and is differentiated by different cycle time p_j^l for producing the products of order j . In addition, the workstations used for the production of order j are indexed by $i \in \mathbb{I}_j^l$, where $\mathbb{I}_j^l$ is a subset of $\mathbb{I}^l$. In the system, AGVs with limited load capacity are used for material delivery with a determined transportation time $T = \{\tau_{l'l'}^{li}, \forall l, l', i, i'\}$, where $\tau_{l'l'}^{li}$ is the total transportation time for an AGV travelling from workstation i of flowline l to the material warehouse plus that from the material warehouse to workstation i' of flowline l' . The material delivery is organised with 'complete kit' as the smallest unit, and a 'complete kit' refers to all the materials required for processing a product on a workstation, always including different types of materials. Due to the limited capacity of AGVs and the different physical characteristics of different materials, the maximum quantity of 'complete kit' that can be carried by a delivery differs between different products and different workstations for the same products. Therefore, the number of delivery tasks carrying materials for producing order j on workstation i of flowline l is denoted as K_{ij}^l . Each delivery task is denoted as T_{lijk} , correspondingly, the production subtask consuming materials delivered by T_{lijk} is denoted as O_{lijk} . Figure 1 shows a simple example of an MFFL production system with AGVs carrying out the in-plant material delivery for 2 flowlines.

To make the MFFL production system perform more efficiently, the ISPMDDP is formulated to decide the assignment and sequence of orders processed on each flowline as well as the assignment and delivery sequence of material delivery tasks. Efficiency and cost-effectiveness are two main objectives of an enterprise's operations management (Tangen 2005). Basically, it is critical for manufacturers to make scheduling decisions to minimise order delays for higher customer satisfaction and stronger market competitiveness (Chiang, Guo, and Pai 2008). In addition, it is also important to reduce the production and operation costs. In an intelligent production system with multiple AGVs, how to use fewer AGVs to meet production requirements and thus reduce costs relating to equipment depreciation and maintenance is another concern (Choobineh, Asef-Vaziri, and Huang 2012). Therefore, we study the integrated production and material delivery scheduling method for minimising total order delays and the number of used AGVs. The notations used to describe the problem are listed in Table 1.

Table 1. Notations.

Indices	
i, i'	Index of workstations
j, j'	Index of orders
k	Index of delivery tasks
l	Index of flowlines
o	Index of the warehouse
Sets	
$\mathbb{J} = \{0, 1, \dots, J, J+1\}$	Set of orders, where $j = 0$ and $j = J+1$ are the dummy start and end orders of a production sequence, respectively
$\mathbb{L} = \{1, \dots, L\}$	Set of flowlines
$\mathbb{I}^l = \{1, \dots, I^l\}$	Set of workstations on flowline l
$\mathbb{I}_j^l \subseteq \mathbb{I}^l$	Set of workstations required when processing order j on flowline l
$\mathbb{K}_{ij}^l = \{1, \dots, K_{ij}^l\}$	Set of delivery tasks required when processing order j on workstation i of flowline l
Parameters	
Q_j	Quantity of products in order j
L	Number of flowlines
H	Number of equipped AGVs
I_j^l	Number of workstations used when processing order j on flowline l
p_j^l	Production cycle time when processing order j on flowline l
d_j	Due date of order j
q_{lijk}	Number of products that can be processed using complete kits of materials delivered by task T_{lijk}
K_{ij}^l	Number of delivery tasks required when order j processing on workstation i of flowline l
$\tau_{l'l'}^{li}$	Total transportation time for AGV travelling from workstation i of flowline l to warehouse plus that from warehouse to workstation i' of flowline l'
$e_{jj'}^l$	Switch time of flowline l when changing from the processing of order j to order j'
Decision variables	
h	Integer variable, number of used AGVs
z_{jl}	Binary variable, equals 1 if order j is processed on flowline l ; 0 otherwise
$x_{jj'}$	Binary variable, equals 1 if the production of order j immediately precedes order j' , 0 otherwise
$y_{l'i'j'k'}^{lijk}$	Binary variable, equals 1 if the delivery task T_{lijk} immediately precedes task $T_{l'i'j'k'}$; 0 otherwise
y_{lijk}^o	Binary variable, equals 1 if the delivery task T_{lijk} is the first task of an AGV delivery route; 0 otherwise
y_o^{lijk}	Binary variable, equals 1 if the delivery task T_{lijk} is the last task of an AGV delivery route; 0 otherwise
t_{lijk}^d	Continuous variable, finish time of a delivery task T_{lijk}
t_{lijk}^f	Continuous variable, finish time of production task O_{lijk}

The following assumptions are made throughout the paper.

- (1) The production of each order has to be completed on a flowline uninterruptedly.
- (2) Each AGV can only transport the material of one workstation each time, and the buffer is large enough.
- (3) The transportation time of the WIP between any two sequential workstations by the conveyor belt is equal to the cycle time.

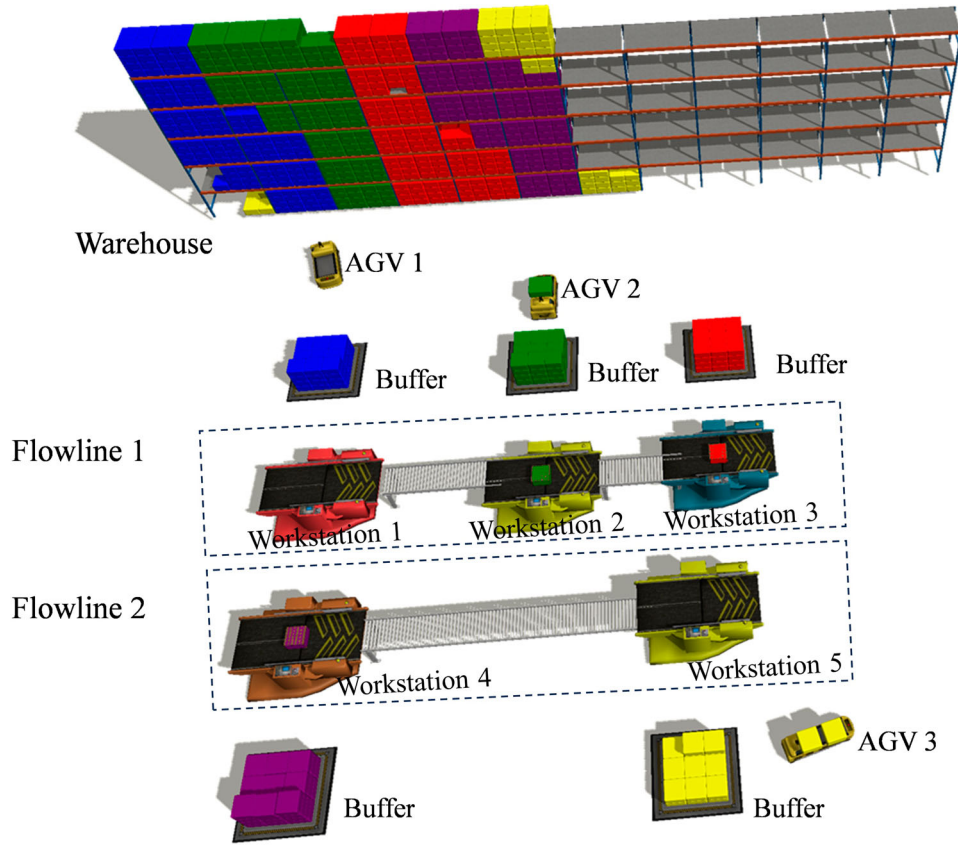


Figure 1. Schematic diagram of an MFFL production system.

3.2. Mathematical model

The ISPMDP is formulated as an MIP model, where one integer variable (h), five sets of binary variables ($z_{jl}, x_{jj'}, y_{l'j'k'}^{lijk}, y_{o}^{lijk}, y_o^{lijk}$), and two sets of continuous variables (t_{lijk}^d, t_{lijk}^f) are used. The production schedules of multiple flowlines are represented by the sequences of several orders, and similarly, the material delivery schedules of the AGVs are represented by the sequences of several delivery tasks.

$$\min \theta_1 \frac{\sum_{j \in J} \max(T_j - d_j, 0) - U_{\min}}{U_{\max} - U_{\min}} + \theta_2 \frac{h - 1}{H - 1} \quad (1)$$

$$\text{s.t. } \sum_{l \in L} z_{jl} = 1, \forall j \in J \quad (2)$$

$$\sum_{j \in J} x_{0j} \leq L \quad (3)$$

$$\sum_{j \in J} x_{j,J+1} = \sum_{j \in J} x_{0j} \quad (4)$$

$$\sum_{j \in J \cup \{0\}} x_{jj'} = 1, \forall j' \in J \quad (5)$$

$$\sum_{j' \in J \cup \{J+1\}} x_{jj'} = 1, \forall j \in J \quad (6)$$

$$x_{jj} = 0, \forall j \in J \quad (7)$$

$$x_{jj'} \leq \sum_{l \in L} z_{jl} z_{j'l}, \forall j, j' \in J \quad (8)$$

$$\sum_{l \in L} \sum_{j \in J} \sum_{i \in I_j^l} \sum_{k \in K_{ij}^l} y_{lijk}^o = h \quad (9)$$

$$y_{lijk}^{lijk} = 0, \forall l \in L, j \in J, i \in I_j^l, k \in K_{ij}^l \quad (10)$$

$$\sum_{l \in L} \sum_{j \in J} \sum_{i \in I_j^l} \sum_{k \in K_{ij}^l} y_{lvirjrk'}^{lijk} + y_{lvirjrk'}^o = z_{jl}, \forall l \in L, j' \in J, i' \in I_{j'}^{l'}, k' \in K_{ij'}^{l'} \quad (11)$$

$$\sum_{l \in L} \sum_{j' \in J} \sum_{i' \in I_{j'}^{l'}} \sum_{k' \in K_{ij'}^{l'}} y_{lvirjrk'}^{lijk} + y_o^{lijk} = z_{jl}, \forall l \in L, j \in J, i \in I_j^l, k \in K_{ij}^l \quad (12)$$

$$2y_{lvirjrk'}^{lijk} \leq z_{jl} + z_{j'l}, \forall l \in L, j, j' \in J, i \in I_j^l, i' \in I_{j'}^{l'}, k \in K_{ij}^l, k' \in K_{ij'}^{l'} \quad (13)$$

$$y_{lijk}^o \leq z_{jl}, \forall l \in L, j \in J, i \in I_j^l, k \in K_{ij}^l \quad (14)$$

$$y_o^{lijk} \leq z_{jl}, \forall l \in L, j \in J, i \in I_j^l, k \in K_{ij}^l \quad (15)$$

$$t_{l'ij'k'}^d \geq \tau_{l'ir}^o y_{l'ij'k'}^o + \sum_{l \in L} \sum_{j \in J} \sum_{i \in I_j^l} \sum_{k \in K_{ij}^l} (t_{lijk}^d + \tau_{lir}^l) y_{l'ij'k'}^{lijk}, \quad (16)$$

$$\forall l' \in L, j' \in J, i' \in I_{j'}^{l'}, k' \in K_{ij'}^{l'}$$

$$t_{l'1j'1}^f \geq \max \left\{ x_{0j'} e_{0j'}^{l'} + \sum_{j \in J} x_{jj'} (T_j + e_{jj'}^{l'}), t_{l'1j'1}^d \right\} + p_{j'}^l q_{l'1j'1} + \Omega(z_{jl} - 1), \forall l' \in L, j' \in J \quad (17)$$

$$t_{lij1}^f \geq \max \{ t_{l,i-1,j,K_{i-1,j}^l}^f - p_j^l Q_j + p_j^l, t_{lij1}^d \} + p_j^l q_{lij1} + \Omega(z_{jl} - 1), \forall l \in L, j \in J, i \in I_j^l \setminus \{1\} \quad (18)$$

$$t_{lijk}^f \geq \max \{ t_{l,i,j,k-1}^f, t_{lijk}^d \} + p_j^l q_{lijk} + \Omega(z_{jl} - 1), \forall l \in L, j \in J, i \in I_j^l, k \in K_{ij}^l \setminus \{1\} \quad (19)$$

$$T_j \geq \sum_{l \in L} t_{ll'jK_{lj}^{l'}}^f, \forall j \in J \quad (20)$$

$$1 \leq h \leq H, h \in Z \quad (21)$$

$$x_{jj'}, x_{0j'}, x_{j,J+1} \in \{0, 1\}, \forall j, j' \in J \quad (22)$$

$$y_{l'ij'k'}^{lijk}, y_{l'ij'k'}^o, y_o^{lijk}, z_{jl} \in \{0, 1\}, t_{lijk}^d, t_{lijk}^f \geq 0, \forall l \in L, j, j' \in J, i \in I_j^l, i' \in I_{j'}^{l'}, k \in K_{ij}^l, k' \in K_{ij'}^{l'} \quad (23)$$

The objective function (1) represents a summation of the normalised value of the two original objectives (Qu and Suganthan 2010), aiming at minimising the order delays and the number of used AGVs simultaneously. The parameter U_{min} represents a lower bound of the order delays, which can be calculated by solving the production scheduling model (the detailed model can be found in Appendix 1). Similarly, U_{max} represents an upper bound of the order delays, which can be set as the objective of a feasible solution. An intuitive method to generate a feasible solution is to schedule the production orders according to the earliest due date rule and schedule the material delivery tasks according to their expected completion time. Similarly, in the second term, scalars 1 and H are the lower bound and the upper bound of the number of used AGVs, respectively. The two nonnegative parameters θ_1 and θ_2 , satisfying $\theta_1 + \theta_2 = 1$, represent the weights of the two objectives, respectively.

Constraints (2)-(8) are about the assignment and sequencing of production orders to MFFL. Constraint (2) guarantees each order is processed on a flowline. Constraint (3) limits the usage of the dummy order $j = 0$ to be less than L , which means the number of flowlines used cannot exceed the number of equipped flowlines.

Constraint (4) ensures the usage of dummy order $j = 0$ is equal to order $j = J + 1$, meaning that the production sequence starts from one order and ends at another. Constraints (5)-(7) describe the sequential relationships of order j and j' . Constraint (8) ensures that two orders can be processed sequentially only if they are allocated to the same flowline.

Constraints (9)-(15) are about the assignment and sequencing of the delivery tasks to the fleet of AGVs. Constraints (9)-(12) describe the sequential relations of delivery tasks of each AGV, whereas constraints (13)-(15) ensure that the delivery task T_{lijk} exists on condition that order j is produced on flowline l , that is $z_{jl} = 1$.

Constraints (16)-(19) calculate the start and completion time of delivery tasks and production tasks. Constraint (16) ensures that the completion time of one delivery task cannot be earlier than that of its immediate predecessor task plus the transportation time between corresponding two workstations. Constraints (17)-(19) calculate the production completion time. Ω is a big enough parameter satisfying $\Omega \geq \max\{T_j, \forall j \in J\}$ and the term $\Omega(z_{jl} - 1)$ on the right side means production task O_{lijk} exists only when order j is allocated to flowline l . Constraint (20) calculates the completion time of order j , which equals the completion time of the last production task of the order. Constraints (21)-(23) are decision variable constraints.

The objective function (1) and constraints (8) and (16)-(19) are nonlinear. We can linearise them by introducing some auxiliary variables, which are shown in Appendix 2. However, the linearised model is still challenging to solve when the number of orders and delivery tasks is large. Therefore, we propose a bi-level optimisation algorithm in the next section to address the complicated problem.

4. Bi-level optimisation algorithm

The ISPMDDP is a confluence of two inherently challenging subproblems, i.e. the MFFL production scheduling problem and the material delivery scheduling problem. The coupling temporal relationship between production and delivery tasks further amplifies the complexity. Therefore, we propose a bi-level heuristics algorithm to solve the integrated scheduling problem.

The procedure of the proposed bi-level algorithm is shown in Figure 2. The primal procedure of the upper-level VNS is illustrated on the left side, and the lower-level ALNS is depicted on the right side. The basic idea of this bi-level optimisation algorithm is to iteratively evolve the production schedule at the upper level, optimise the material delivery schedule while adhering to the given order sequences at the lower level, and subsequently feed

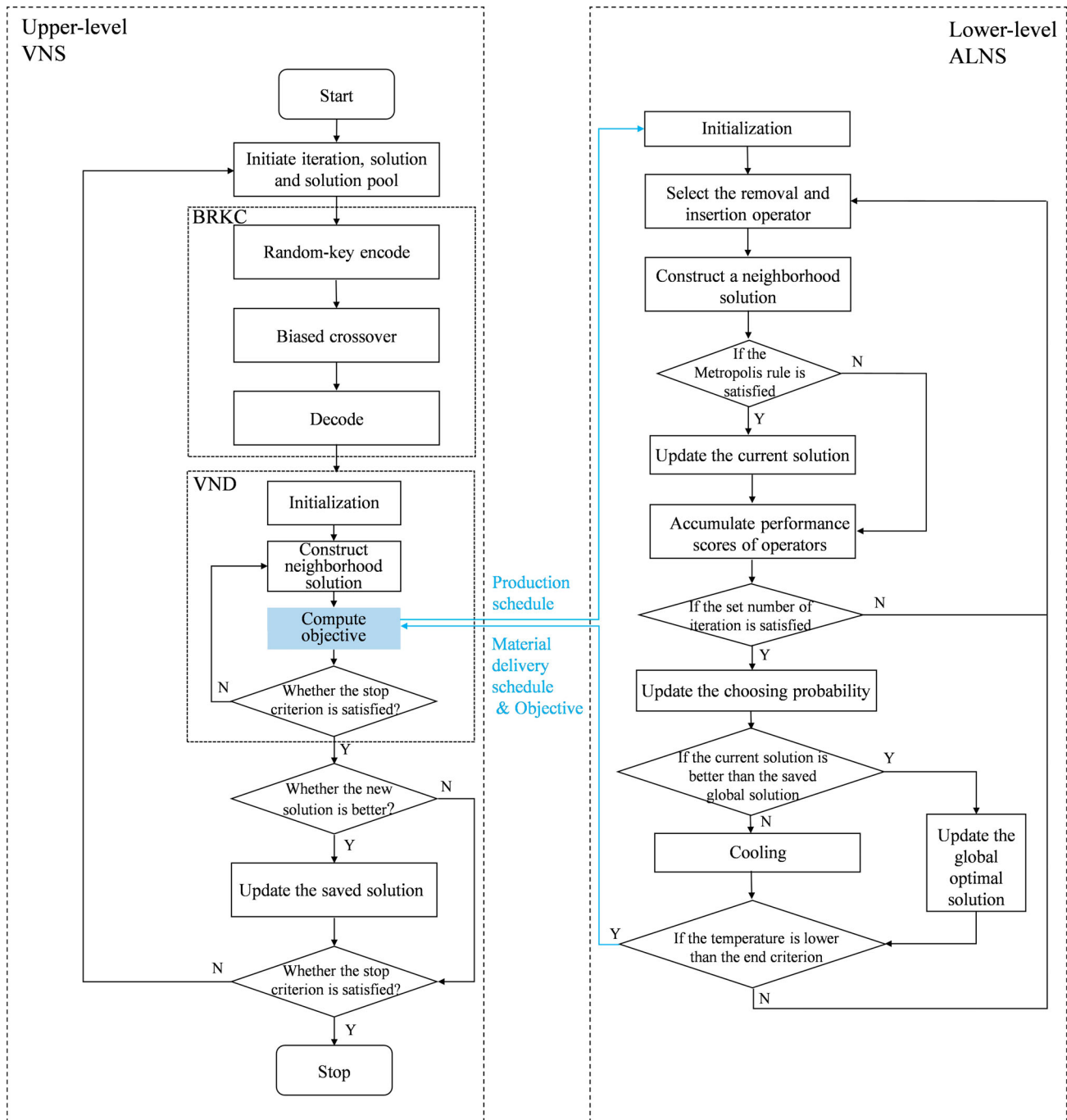


Figure 2. A bi-level optimisation algorithm.

the objectives back to the upper level for further iterative optimisation.

In the upper level, the algorithm focuses on optimising the assignment of orders to flowlines and determining the production sequence on each flowline. The algorithm commences with the generation of a random initial solution. During each iteration, we employ the BRKCopoperator to shake the current solution, thereby enhancing its ability to evade local optima. After that, the solution is decoded, and a variable neighbourhood

descent (VND) procedure is applied, within which the ALNS (depicted in blue) is nested for an intensive local search. The algorithm iterates until the stopping criterion is met.

In the lower-level ALNS algorithm, the primary focus revolves around optimising the number of AGVs in use and the schedule of material delivery tasks. The process initiates with the creation of an initial material delivery schedule according to the expected arrival time of delivery tasks given the input production schedule. To

maintain an effective and adaptive search process, the algorithm operates within a simulated annealing (SA) scheme. In each iteration, one removal and one insertion operator are selected to construct a neighbourhood solution according to their historical performances in improving the quality and diversity of solutions, making it possible to modify the search process adaptively.

4.1. Upper-level VNS

The VNS framework employed at the upper level was originally conceptualised by Mladenović and Hansen (1997) and has garnered widespread adoption across diverse fields within the realm of operations research. Notably, it has been effectively employed in tasks such as clustering, VRP, and scheduling problems (Gendreau and Potvin 2019). The VNS algorithm hinges on two pivotal mechanisms: random shaking and neighbourhood search, which play crucial roles in local optima avoidance and local optimisation, respectively. In our VNS algorithm, these two mechanisms are realised through a BRKC operator and a VND approach with five distinct neighbourhood construction operators.

4.1.1. BRKC

Building upon the random-key encoding method introduced by Soares et al. (2020), this paper incorporates a BRKC operator to shake the newly constructed neighbourhood solution. The main idea of the BRKC operator is to perform a parameterised two-point crossover between the encoded solution (referred to as an individual) and an individual randomly selected from the individual pool. The newly generated child solution is decoded, and the local search process proceeds.

The random-key encoding and decoding method is employed to circumvent potential computational complexities arising from the need to rectify infeasible child solutions. Given J orders to be produced on L flowlines, the production schedule can be denoted by L order sequences, each encompassing $J_l (l = 1, \dots, L)$ orders, satisfying the condition $\sum_{l \in \mathbb{J}} J_l = J$. To encode a production schedule, J_l decimals within the range $[l, l + 1)$ are initially generated randomly for each l . Subsequently, the decimals of the l th sequence are sorted in ascending order, and each decimal serves as the key for the order in the corresponding position. The encoded sequences are then connected, aligning them with their l indexes, resulting in a real-number encoded solution.

To control the randomness, the BRKC performs the uniform crossover with the probability parameter a^* . At each position of the incumbent solution, a decimal a within the range $(0, 1)$ is generated and compared with

the given a^* . If a is larger than a^* , the current key is replaced by that of the selected individual; otherwise, the key remains unchanged. With all positions having been scanned, the crossover process is completed and the newly generated offspring is added to the individual pool.

The final step of the BRKC is to decode the offspring. Orders are firstly grouped into at most L sequences according to the integer part of their keys and sorted according to their fractional part. For example, orders with keys 1.28 and 1.33 are allocated to flowline 1 and the order keyed 1.28 is placed before the order keyed 1.33. Then, a shaken production sequence is translated based on the key-order relationship. Since each real number in an encoded solution serves as the key to a unique task, and the crossover process is operated between two keys of the same task, a newly generated solution is still a permutation sequence of all delivery tasks, i.e. a feasible solution. An illustrative example of the BRKC is provided in Appendix 3.

4.1.2. VND

This paper employs the VND method as a systematic approach for conducting local search operations across various neighbourhoods to approximate the global optimum to a reasonable extent (Gendreau and Potvin 2019). The primary steps of the VND are as follows: Initially, the status parameter η is set to 1. In each iteration, a local search is executed within the η^{th} neighbourhood, which is centred around the incumbent solution p' , resulting in a new solution denoted as p'' . If the objective function value of the new solution p'' is superior to that of the incumbent solution, the VND process restarts from p'' ; otherwise, the status parameter is incremented by 1, and the neighbourhood search process continues. The VND process ends when all neighbourhoods have been sequentially explored once without any restarts, indicating that the incumbent solution is indeed optimal within all the searched neighbourhoods.

Our VND includes five neighbourhood construction operators. Aiming at balancing the workload of flowlines, the *order insertion* operator attempts to remove the least time-consuming order from the flowline with the longest processing time ($l_{\max}$) and insert it in the flowline with the shortest processing time ($l_{\min}$). The *order exchange* operator is geared toward exchanging the most time-consuming order from $l_{\max}$ and the least time-consuming order from $l_{\min}$. Both of these operators draw inspiration from the work of Leonardo C.R. Soares and Carvalho (2022). In the following, we propose three other operators that are tailored to the distinctive characteristics of ISPM DP.

4.1.2.1. Logistics balance operator. Considering a production schedule where the material requirements of most flowlines are centralised at a certain time, the execution of the production schedule may be interrupted due to the limited logistics resources. This operator is specifically devised to address this issue by balancing the material delivery workload across all flowlines. To facilitate clarity, we introduce the variable idx to represent the position of an order within a production sequence. The operator quantifies the delivery task density for order j , which is defined as the sum of the number of delivery tasks required for its processing across all workstations, denoted as $TD_j = \sum_{i \in \mathbb{I}_j} 1/c_{ij}p_j$. Furthermore, we establish

the concept of delivery task density for a specific position, which is the cumulative density of all orders on that position of all flowlines ($\mathbb{O}_{idx}$), denoted as $TD_{idx} = \sum_{j \in \mathbb{I}} TD_{idx,j}$. The detailed steps are as follows. Initially, it calculates the material delivery task density for all positions and selects the one with the highest density (position idx). Subsequently, it computes the slack time for each order in $\mathbb{O}_{idx}$ by calculating $d_j - T_j$. To balance the logistics burden, the order with the longest slack time is exchanged with one preceding order j' from the same production sequence. Order j' have to meet the condition that its delivery task density is smaller than order j , and at the same time, its slack time is larger.

4.1.2.2. Difference reduction operator. In general, given a schedule of orders and material delivery tasks, the scheduled processing time ($Q_j * p_j^l$) and actual processing time ($T_j^f - T_{j'}^f + e_{j,j}^l$) of an order can be calculated easily. A greater time difference suggests additional time spent beyond the manufacturing process, such as waiting for materials. To improve the current schedule, a possible direction is to reallocate orders with a large time difference to a more appropriate position. Based on this idea, this operator first calculates the difference between the scheduled and actual production time of all orders. After that, an order j' is randomly selected from the group of orders with the $1/\varepsilon$ smallest differences, where ε is a given parameter. The last step is to exchange the position of order j with the largest time difference and order j' to construct a new production schedule.

4.1.2.3. Slack reduction operator. Considering a production sequence with zero order delay, some orders may be found to have large enough slack time for inserting orders that are originally allocated to another sequence into this one. Consequently, the completion time of the relocated orders may be brought forward,

leading to potential improvements in the overall objective. This operator first calculates delays of all orders ($|T_j^f - d_j|$) and the total slack time of orders in each production sequence $\left(\sum_{j \in \mathbb{I}_l} (d_j - T_j^f) \right)$; then, the order with the longest delays is denoted as order j , and the production sequence with the longest total slack time is denoted as l_{max} . The next step is to insert order j into the production sequence l_{max} , and the insertion happens immediately after the first order with an earlier due date than order j in an inverted order.

4.2. Lower-level ALNS

The ALNS framework was first developed by Ropke and Pisinger (2006), which inherits the concept of neighbourhood construction that uses removal and insertion operators to damage and repair the current solution and generate a new feasible one (Shaw 1998). Besides, mechanisms for adaptive operator selection and SA-based varied solution acceptance are introduced to enhance its global search ability. The initial solution construction and the removal and insertion operators in our algorithm are detailed in the following.

4.2.1. Initial solution generation

The lower-level subproblem solved by ALNS is a specialised VRP where the nodes to be visited refer to the material delivery tasks, and the routes of a vehicle correspond to the task sequences. To generate the initial solution, we first divide material requirements into several delivery tasks based on the AGV's load capacity and then, calculate the expected completion time with the given cycle time and production quantity of each task. All delivery tasks are assigned in ascending order of expected completion time, and two consecutive tasks are allocated to two routes with adjacent indexes. This method aims to achieve a balance in the number of delivery tasks allocated to each AGV while maintaining the temporal sequence based on the expected completion time of all tasks on each route.

4.2.2. Removal operators

Introducing variations to some variable values in the current solution is a method employed by ALNS to construct a neighbourhood solution. In this subsection, we present some removal operators to establish rules for removing and reallocating delivery tasks. We utilise *random removal* and *worst removal* with fundamental concepts inspired by Pisinger and Ropke (2007). Additionally, we introduce the *time difference removal* and *route cut removal* operators. Following the approach suggested by

Pisinger and Ropke (2007), this paper generates a random integer, denoted as q , to determine the number of nodes to be removed in each iteration, preventing repetitive reallocation of specific tasks.

4.2.2.1. Time difference removal operator. Similar to the concept of the *difference reduction* operator, this operator aims to reallocate the material delivery tasks with a significant time difference. In addition, since the ISPM-DP assumes a fixed process flow for each product, delays in material delivery at the preceding workstation may defer production at subsequent workstations. Delays in preceding orders may also postpone the production of successive orders, given the nature of MTO production. Therefore, among all delivery tasks, denoted as $\{u\}$, those with an earlier scheduled completion time ($time_u^s$) should be given higher removal priority to mitigate cumulative impacts on other production tasks. Hence, the *time difference removal* operator assesses the removal priority of each delivery task using an evaluation index $d_u = \mu * (time_u^d - time_u^s) - (1 - \mu) * time_u^s$, where parameter $\mu \in (0, 1)$ denotes the weight of the time difference in the overall evaluation.

4.2.2.2. Route cut removal operator. To avoid logistics congestion in the production system, the route cut removal operator is introduced to reduce the number of AGVs used. If a superior solution with fewer AGVs in use is found, the neighbourhood search process continues with fewer AGVs; otherwise, the number of AGVs used reverts to the previous status. This operator introduces the potential for achieving a trade-off between the two optimisation goals.

4.2.3. Insertion operators

Next, we introduce the insertion operator designed to repair the incomplete solution that has been ‘damaged’ by removal operators. This set of operators inserts tasks from the removal set into new positions within existing task sequences. Four insertion operators are utilised. The *random insertion* operator inserts tasks into random positions. The *greedy insertion* operator places tasks in positions with the least objective deterioration. The remaining two operators, namely, the *partial greedy insertion* operator and the *time order insertion* operator, are proposed to address the computational complexity associated with the greedy approach.

4.2.3.1. Partial greedy insertion operator. Inspired by the simple insertion operator proposed by Ben Ticha et al. (2019), this operator combines the ideas of *random insertion* and *greedy insertion*. By randomly choosing an existing sequence, the feasible region for

insertion enumeration is significantly reduced, facilitating the subsequent greedy search on the selected sequence. To implement the idea, the task at the front of the removal task set is selected and removed first. Then, a delivery sequence from the incomplete solution is randomly chosen. All positions in the selected sequence are enumerated and the objective after inserting is calculated. Each task is inserted into the position resulting in the minimum objective. The operator terminates when all tasks in the removal set have been inserted into the schedule.

4.2.3.2. Time order insertion operator. This operator enhances the greedy insertion by contracting a task’s insertion substitution space to its reasonable positions, denoted as $\{rp_u^r, \forall u, r\}$. For example, in delivery sequence r , $rp_u^r = v + 1$ means the scheduled completion time of task u is later than that of the v^{th} task, while earlier than that of tasks from the $(v + 2)^{\text{th}}$ position to the end of sequence r . The steps are as follows. First, select and remove a task u from the removal task set; then, locate the rp_u^r in each sequence r ; later, compute the objectives with the removal task inserted in each rp_u^r ; finally, insert the task in the position with the minimum objective after its insertion. The heuristic terminates when all removal tasks are reinserted.

The *time difference removal* operator, as discussed in section 4.2.2, considers the reallocation priority of delivery tasks based on some temporal criterion, necessitating the reinsertion of these tasks following the temporal rules. Hence, this paper binds the *time order insertion* operator to the *time order removal* operator to strengthen the algorithm’s pertinence. Meanwhile, due to the generality of the *time order insertion* operator, it can be used with other removal operators.

5. Numerical experiments

In this section, we conduct extensive numerical experiments with real-case-based instances from a well-known air conditioning compressor manufacturer, to evaluate the performance of our proposed approach under various parameter settings and problem scales. All programmes are developed using Python 3.10 and the MILP models are solved through the commercial solver Gurobi 9.5.0. All experiments are implemented on a workstation configured with [Intel Xeon W-2245 CPU (3.90 GHz, 8 cores), 64GB RAM]. Section 5.1 details the settings of our numerical experiments. Section 5.2 finds out the superiority of making integrated schedules under diverse parameter settings, and Section 5.3 performs experiments with various problem scales. Discussions on

Table 2. Settings in the base case (D0).

Order				
Index	Product	Quantity (Unit)	Cycle time (s)	Due time (s)
A	a	200	8	6000
B	b	100	9	3768
C	c	100	11	5512
D	d	50	9	6598
Delivery time				
(s)(from/to)	Warehouse	Workstation1	Workstation2	
	0	600	650	
Workstation1	600	1100	1150	
Workstation2	650	1150	1200	
Mould switch time				
(s)(from/to)	a	b	c	d
	300	400	200	600
a	0	200	400	500
b	100	0	300	300
c	300	400	0	200
d	300	200	400	0
Number of products per task				
(Unit)(Workstation/Order)	A	B	C	D
1	50	50	100	50
2	40	100	100	50

Table 3. Parameters of instances D1 ~ D18.

Instance index	L	J	H	K_{ij}
D1 ~ D3	2	4	2, 3, 5	10
D4 ~ D6	2	4	3, 4, 5	10
D7 ~ D9	2	6	5, 8, 10	50
D10 ~ D12	3	10	5, 8, 10	50
D13 ~ D15	4	20	5, 8, 10	100
D16 ~ D18	6	26	5, 8, 10	150

the managerial insights are presented at the end of this section.

5.1. Experiment setting

We construct numerical instances based on real data, including customer orders, factory layout, cycle time, bill of materials (BOM), and other relevant data sourced from our collaborative manufacturer. We establish a base case (D0) for the performance comparison experiments detailed in Section 5.2 as follows (Table 2).

In addition, we generate another 18 instances, denoted as D1 ~ D18, to test the proposed algorithm under different problem scales in Section 5.3. Parameters such as the number of production lines L , the number of orders J , the number of quipped AGVs H , and the number of delivery tasks K_{ij} are listed in Table 3. (Detailed data of each instance can be found in online supplemental materials.)

The quantity of products in each order is set as an integer within [50, 1000]. The cycle time of each type of product is set as an integer within [8, 15]. We consider the heterogeneous flowlines by distinguishing the cycle time of products on each flowline. The time to change moulds to produce different products is set to less than 1800 s.

The parameters used in the bi-level algorithm are obtained through numerical experiments by comparing different parameter values. Initially, the values are set with reference to the literature (Ropke and Pisinger 2006), and then tuned one by one to balance the run-time and solution quality. The parameters are finally set as Table 4.

5.2. Performance comparison under different parameter settings

In this subsection, we compare the solutions derived from integrated and sequential scheduling methods, with the primary aim of verifying the necessity of integrated production and material delivery scheduling in the context of intelligent flowline factories and analysing the influence of problem parameters on the performance of different scheduling methods. Three methods are compared. The first is an integrated scheduling approach that leverages Gurobi to solve the model presented in Section 3, denoted as ISG. The second and third methods are sequential scheduling methods, which are denoted as SSG and SSGP, respectively. The SSG method uses Gurobi to solve the production scheduling problem and the material delivery scheduling problem sequentially, whereas the SSGP method uses Gurobi to solve the production scheduling problem and apply the first-come-first-served rule for the material delivery scheduling. The details regarding the models used in SSG and SSGP are provided in Appendix 1. This subsection performs sensitivity analyses concerning four critical factors, i.e. the number of configured AGVs, delivery time, the number of orders, and the weightings assigned to objectives.

5.2.1. Analyses of the number of configured AGVs

We vary the number of configured AGVs from 3 to 7. The results are shown in Table 5. One can see that the ISG method gets the best objective values under all settings, followed by SSG and SSGP. Note that some objectives obtained by SSG and SSGP are larger than 1, which is because the delays of these solutions are larger than the U_{max} set in section 3.2. Such results can also be observed in the following experiments, e.g. Tables 6–8. In terms of the number of used AGVs, the SSG method always uses the maximum, whereas the ISG method uses all AGVs when the configured AGVs are relatively few and uses the fewest AGVs when more AGVs are available. Additionally, in terms of order delays, the ISG method achieves the least delays. With the increase of the configured AGVs, the ratios of improvements on delay minimisation by the three methods are 100%, 82%, and 23%, respectively. The results indicate that compared with the two sequential methods, the integrated scheduling method makes

Table 4. Parameter setting.

Parameter	Value	Interpretation
w	0.002	Initial temperature
α	0.995	Cooling coefficient
L_T	100	Iterations under a temperature
L	1000	Total iterations of the ALNS
$\sigma_1, \sigma_2, \sigma_3$	9,2,1	Scores of neighbourhood search operators gained in each iteration due to the quality of solutions they constructed
ρ	0.15	Weight of the performance under the current temperature in the total evaluation
β	6	The randomised coefficient for removing tasks from a sorted list
$\delta_1, \delta_2, \delta_3$	0.15,0.06,0.3	The upper limit of the proportion of removal tasks to total tasks by the random removal, worst removal, and time difference removal operators, respectively
μ	0.59	Weight of the difference between the actual and expected completion time in the synthetical index of the time difference removal operator
l_{min}	100	The smallest iteration number of VNS
l_{Δ}	10	The largest iteration number without improvement of VNS
$bPro$	0.8	Probability of inheriting the current solution in the crossover shaking operator of VNS

Table 5. Performance of different methods with respect to (w.r.t.) the number of configured AGVs.

Number of configured AGV	ISG			SSG			SSGP		
	Objective	Used AGV	Order delay (s)	Objective	Used AGV	Order delay (s)	Objective	Used AGV	Order delay (s)
3	0.80	3	2838	1.16	3	6180	1.84	2	14927
4	0.61	4	958	0.96	4	3988	1.90	2	14927
5	0.54	5	126	1.12	5	2106	4.33	4	13388
6	0.45	5	126	1.31	6	1870	5.63	6	11820
7	0.42	6	0	1.26	7	1124	8.24	7	11477

more use of existing in-plant logistics resources so that it can improve production efficiency when more AGVs are configured in the system.

5.2.2. Analyses of the delivery time

We consider the impacts of the delivery time by varying the delivery time from the warehouse to workstation 1 and to workstation 2. In fact, the average length of material delivery time per task is a decisive factor in determining the relative abundance of in-plant logistics resources, such as AGVs. The longer the average delivery time, the greater the shortage of AGVs. Table 6 shows the solutions of the three methods under various delivery time settings. It can be observed that our proposed integrated scheduling method outperforms the two sequential scheduling methods in terms of the objective value. Additionally, in terms of the order delay, when the delivery time is (100, 150), all methods can achieve zero delay. Specifically, when delivery time is short and AGVs are abundant, ISG uses fewer AGVs to perform on-time material delivery compared to the other two sequential scheduling methods. As delivery time increases, all three methods use the maximum number of AGVs due to the increasing pressure on material delivery. However, ISG still achieves shorter order delays by distributing the material requirements more evenly over time.

5.2.3. Analyses of the number of orders

This subsection analyses the impacts of the number of orders for various solution methods. Based on D0, we

construct instances by varying the number of orders from 2 to 5 and correspondingly increasing delivery tasks. The 2-order and 3-order instances are generated by extracting the first 2 and 3 orders of the base case, and the 5-order example is generated by adding a new order. At the same time, the number of delivery tasks are controlled to increase with the growing order number. Table 7 shows the solution results. Through horizontal comparison, we find that although ISG yields better objectives under most parameter settings, its superiority is particularly significant under the 4-order case. A possible explanation is that, as the number of orders increases, the relative shortage of AGV resources limits the potential to mitigate the unpunctuality of material delivery through carefully integrated scheduling. Furthermore, the objectives obtained by ISG show a steady increasing trend, while the objective values obtained by SSG and SSGP fluctuate irregularly. This counterintuitive phenomenon highlights one reason for applying the integrated scheduling method: to hedge against deviations caused by the random adoption of non-unique, equally optimal production schedules in the first step of a sequential scheduling method.

5.2.4. Analyses of the objective weightings

Next, we analyse the impacts of the objective weightings by varying parameter θ_1 from 0.1 to 1. In Table 8, we can also see that the integrated scheduling method performs best in all settings. As the parameter θ_1 increases, it can be observed that at first, the number of used AGVs of all methods increases, while the order delay of all methods

Table 6. Performance of different methods w.r.t. delivery time.

Delivery time (s, s)	ISG			SSG			SSGP		
	Objective	Used AGV	Order delay (s)	Objective	Used AGV	Order delay (s)	Objective	Used AGV	Order delay (s)
(100, 150)	0	1	0	0	1	0	0	1	0
(200, 250)	0.13	2	0	0.25	3	0	0.21	2	289
(300, 350)	0.25	3	0	0.50	5	0	0.86	2	2487
(400, 450)	0.38	4	0	0.62	5	407	1.85	4	4988
(500, 550)	0.50	5	0	1.50	5	1106	3.09	4	9188
(600, 650)	0.54	5	126	1.12	5	2106	4.33	4	13388
(700, 750)	0.60	5	626	1.00	5	3106	5.58	4	17588
(800, 850)	0.63	5	1126	0.96	5	4106	7.05	4	22558

Table 7. Performance of different methods w.r.t. order numbers.

Order number	Delivery task number	ISG			SSG			SSGP		
		Objective	Used AGV	Order delay (s)	Objective	Used AGV	Order delay (s)	Objective	Used AGV	Order delay (s)
2	12	0.30	3	0	395.50	5	790	2001.00	5	4001
3	14	0.50	5	0	0.50	5	0	5.84	5	3629
4	16	0.54	5	126	1.12	5	2106	2.44	5	6570
5	22	0.56	5	534	0.58	5	630	2.15	5	13569
6	25	0.64	5	2047	0.63	5	1901	3.12	5	21535

decreases. This is because as θ_1 increases, the weight of minimising the order delay increases, while the weight of minimising the number of used AGVs decreases. However, the effect of reducing order delay by adjusting objective weightings is limited. It can be observed that the order delays for all three scheduling methods remain unchanged when the θ_1 values are 0.6 and higher. This limitation underscores the necessity of integrated scheduling to further improve production efficiency.

In summary, from the above sensitivity analyses, we can see that all four parameters have a significant impact on the solution, and our proposed integrated scheduling method outperforms the two sequential scheduling methods, which demonstrates the value of integrated scheduling. In settings with relatively tight logistics capacity, e.g. fewer AGVs, longer delivery time, the integrated scheduling shows a greater advantage. When the delivery capacity is relatively abundant, the integrated scheduling can achieve on-time material delivery with fewer AGVs. Besides, the increasing order number does not reveal more obvious advantage of integrated scheduling in our experiments, which implies other factors such

as the required delivery task number have to be involved in the selection of scheduling method. Moreover, comparing within the two sequential scheduling methods, the model-based SSG method outperforms the rule-based SSGP method.

5.3. Experiments on problem scales

This section compares the performance of three scheduling methods under various problem scales. The sequential scheduling algorithm (SSA) makes use of the proposed algorithm VNS and ALNS to sequentially establish the production schedule and material delivery schedule. Two methods are used for integrated scheduling, one is the ISG mentioned in section 5.2, and the other is the BOA proposed in section 4. The upper bound for the computation time of each method is limited to 10800 s.

5.3.1. Small-scale experiments

This subsection studies the results of small-scale instances D1 ~ D6. Table 9 shows the objective, gap, number of used AGVs, order delay, and computation time of various solution methods for different instances. The columns

Table 8. Performance of different methods w.r.t. weightings of objectives.

θ_1	ISG			SSG			SSGP		
	Objective	Used AGV	Order delay (s)	Objective	Used AGV	Order delay (s)	Objective	Used AGV	Order delay (s)
0.1	0.57	2	5899	0.82	3	6180	1.11	2	14927
0.2	0.71	4	958	1.05	5	2106	1.97	2	14927
0.3	0.69	4	958	1.07	5	2106	2.82	2	14927
0.4	0.63	5	126	1.10	5	2106	3.62	4	13388
0.5	0.54	5	126	1.12	5	2106	4.33	4	13388
0.6	0.44	5	126	1.15	5	2106	5.03	5	13059
0.7	0.35	5	126	1.17	5	2106	5.71	5	13059
0.8	0.26	5	126	1.20	5	2106	6.39	5	13059
0.9	0.17	5	126	1.22	5	2106	7.05	5	13059
1.0	0.07	5	126	1.25	5	2106	7.72	5	13059

Table 9. Results of small-scale instances.

Instance		D1	D2	D3	D4	D5	D6	Average
SSA	Obj_{SSA}	0.80	0.57	0.40	0.49	0.52	0.41	0.53
	Number of used AGV	2.00	2.00	4.00	2.00	3.00	4.00	2.83
	Order delay (s)	2320.36	2326.60	100.00	2964.12	1717.00	550.00	1663.01
	$Time_{SSA}$ (s)	6.80	6.85	14.78	7.27	10.22	13.97	9.98
ISG	Obj_{ISG}	0.65	0.41	0.40	0.41	0.39	0.36	0.44
	Gap_{ISG}	-18.75	-28.07	0.00	-16.33	-25.00	-12.20	-16.72
	Number of used AGV	2.00	2.00	4.00	2.00	2.00	2.00	2.33
	Order delay (s)	1161	1161	100	2011	2011	2011	1409.17
BOA	$Time_{ISG}$ (s)	385.32	21.27	8.91	157.42	26.29	27.30	104.42
	Obj_{BOA}	0.71	0.47	0.40	0.43	0.43	0.38	0.47
	Gap_{BOA}	-11.25	-17.54	0.00	-12.24	-17.31	-7.32	-10.94
	Number of used AGV	2.00	2.00	4.00	2.00	2.00	2.00	2.33
	Order delay (s)	1599.12	1560.52	100.00	2276.28	2279.32	2172.6	1664.64
	$Time_{BOA}$ (s)	98.16	106.33	199.70	105.87	120.31	180.64	135.17

$Gap_{ISG} = (Obj_{ISG} - Obj_{SSA}) / Obj_{SSA} \times 100$ and $Gap_{BOA} = (Obj_{BOA} - Obj_{SSA}) / Obj_{SSA} \times 100$ are the improved percentages of the two integrated methods (ISG, BOA) compared to the separate method (SSA).

The column Obj_{ISG} is the exact global optimal objective of the ISPMMP, which is 16.72% better than the results of SSA on average. Meanwhile, the results of BOA are 10.94% better than that of the SSA. In terms of running time, the SSA costs 9.98 s, which is the least among the three methods. Although the BOA costs 135.17 s on average, we observe that its objective stops improving after 11.165 s, which suggests setting different maximum iteration times according to the actual needs. In addition, the ISG method has the smallest order delays with an average value of 1409.17 s with the minimum used AGVs. The integrated algorithm BOA performs better than the sequential method SSA for achieving a similar order delay with fewer AGVs.

5.3.2. Medium-scale experiments

The results of instances D7 ~ D12 are shown in Table 10. In terms of the solution objective, we find that the BOA performs best as it achieves the minimum order delays with the least AGVs in use. In terms of the running time, the ISG method reaches the upper bound of 10800 s, indicating that Gurobi cannot get the exact solution of the MILP model in this given time limit. With these medium-scale instances, we can summarise that the performance of the integrated scheduling method is better than the sequential scheduling method, and the BOA outperforms the ISG method.

5.3.3. Large-scale experiments

The results of large-scale instances D13 ~ D18 are displayed in Table 11. The '-' in the table represents that Gurobi cannot find a feasible solution to the MILP model in 10800 s. It can be seen that the BOA algorithm performs best in terms of the solution objective, followed by

SSA and ISG. The ISG algorithm cannot find a feasible solution within 10800 s for instances D16 ~ D18.

In addition, the average order delays of BOA are the smallest among the three methods, with an average value of 34664.09 s, which is only 51.77% of SSA. All the solutions obtained by ISG for instances D13 ~ D15 are the worst, with the most AGVs in use and the largest order delays. From the above results, we can see that the ISG method with Gurobi cannot solve the scheduling problem effectively, while our proposed BOA algorithm performs best.

According to the above experiments, we know that integrated production and material delivery schedules improve the performance of the flowline production systems, resulting in shorter order delays and fewer used AGVs. Therefore, applying an integrated scheduling method is beneficial for manufacturers competing in a fierce market. Specifically, when AGVs are relatively inadequate (either due to a small number of configured AGVs or long delivery times per task), integrated scheduling can smooth in-plant material delivery requirements over time, greatly reducing order delays. It has also been shown that this improvement cannot be replicated by simply increasing investment in in-plant logistics resources. Since the superiority of integrated scheduling becomes more apparent with larger problem scales, offering significant benefits to intelligent manufacturers with large-scale factories producing a variety of products. The experiment results also suggest a limit to the adjustment of objective weights, providing further motivation for manufacturers to adopt integrated scheduling methods.

Although an integrated schedule can be created by solving the MILP model using commercial solvers, its weakness becomes apparent as the problem scale increases. The proposed optimisation algorithm offers a promising alternative for factory managers to handle the integrated scheduling problem of a practical scale. Since the algorithm converges quickly to a good solution,

Table 10. Results of medium-scale instances.

Instance		D7	D8	D9	D10	D11	D12	Average
SSA	Obj_{SSA}	0.75	0.47	0.39	0.70	0.48	0.39	0.53
	Number of used AGV	5.00	7.00	8.00	5.00	6.00	7.00	6.33
	Order delay (s)	9652.08	907.00	0.00	15154.44	5746.20	3068.80	5754.75
	$Time_{SSA}$ (s)	483.91	476.27	486.88	733.38	713.64	740.10	605.70
ISG	Obj_{ISG}	0.80	0.51	0.46	0.55	0.37	0.28	0.50
	Gap_{ISG}	6.40	7.99	18.29	-21.24	-22.66	-27.30	-6.42
	Number of used AGV	5.00	8.00	7.00	5.00	6.00	5.00	6.00
	Order delay (s)	11376.00	459.00	3618.00	3830.00	889.00	3610.00	3963.67
BOA	$Time_{ISG}$ (s)	10800	10800	10800	10800	10800	10800	10800
	Obj_{BOA}	0.67	0.45	0.35	0.43	0.29	0.24	0.41
	Gap_{BOA}	-10.69	-5.13	-8.74	-38.02	-38.52	-36.62	-22.95
	Number of used AGV	5.00	7.00	7.00	4.00	4.28	4.96	5.37
	Order delay (s)	6572.08	600.96	616.00	4426.69	3726.08	1457.58	2899.90
	$Time_{BOA}$ (s)	1690.26	1793.47	1821.61	7884.02	8098.86	8933.32	5036.92

Table 11. Results of large-scale instances.

Instance		D13	D14	D15	D16	D17	D18	Average
SSA	Obj_{SSA}	0.54	0.35	0.30	0.76	0.54	0.45	0.49
	Number of used AGV	4.04	5.02	5.86	5.00	6.00	7.00	5.49
	Order delay (s)	23127.74	8677.63	3888.93	181973.29	114527.14	69586.67	66963.57
	$Time_{SSA}$ (s)	3708.72	4473.00	4269.24	10800	9442.05	10800	7266.95
ISG	Obj_{ISG}	0.84	0.50	0.33	-	-	-	-
	Gap_{ISG}	55.48	43.10	9.92	-	-	-	-
	Number of used AGV	5.00	7.00	7.00	-	-	-	-
	Order delay (s)	48947	8600	0.00	-	-	-	-
BOA	$Time_{ISG}$ (s)	10800	10800	10800	10800	10800	10800	10800
	Obj_{BOA}	0.35	0.22	0.18	0.52	0.37	0.30	0.32
	Gap_{BOA}	-34.82	-35.53	-40.00	-31.43	-31.69	-33.35	-34.47
	Number of used AGV	3.81	4.14	4.27	3.94	5.02	4.95	4.36
	Order delay (s)	0.00	0.00	0.00	107194.25	52280.05	48510.23	34664.09
	$Time_{BOA}$ (s)	10800	10800	10800	10800	10800	10800	10800

practitioners are suggested to select a smaller iteration limit when solving small-scale problems.

6. Conclusion

The rationality of the production schedule is pivotal for efficient production. In the context of intelligent factories utilising AGVs for in-plant material delivery, the interdependence between production and material delivery necessitates integrated scheduling. In this paper, we formulate an MILP model to address the ISPMDDP. Specifically, the interaction of the two considered subproblems – production order scheduling and material delivery scheduling – is thoroughly analysed and incorporated into the temporal coupling constraints. Given its inherent complexity, we propose a bi-level optimisation algorithm based on neighbourhood search. In addition, new operators are devised for both algorithm levels to enhance their local search capability. Sensitivity analyses of problem parameters and scales are conducted through extensive numerical experiments based on data from our collaboration with an air conditioning compressor manufacturer. The results indicate that integrated

scheduling is beneficial in reducing both order delays and the number of AGVs used, especially when the problem scale is large, which is common in intelligent manufacturing systems. Furthermore, the smoothing effect of integrated scheduling on material delivery requirements cannot be replicated by simply adding hardware investment or increasing the weights of certain optimisation goals. The experiments also verify that our proposed bi-level heuristics efficiently create integrated schedules, with objectives 19.28% better than those obtained by Gurobi in a relatively shorter computation time. Additionally, our algorithm can generate acceptable feasible schedules when Gurobi fails.

The model and algorithm proposed in this paper can be applied in a wide range of industries, such as household appliances, electronic products, and cosmetics. Our research demonstrates that the proposed BOA is an effective method for implementing integrated scheduling in practical production scenarios, either by integrating it with existing production management systems, such as the advanced planning and scheduling (APS) system or by using it independently as a tool. Since the parameter values provided in this paper are based on our

studied instances, manufacturers utilising this heuristic algorithm should consider these parameters carefully. For example, the maximum number of iterations can be adjusted to achieve a balance between the running time and the solution quality. In addition, the BOA provides a compatible framework for further personalised modifications. Managers can design new neighbourhood solution construction operators and incorporate them into the current BOA scheme. The flexibility stems from the adaptive mechanism, wherein several embedded neighbourhood solution construction operators are adaptively selected during iterations. When a new operator is added to the framework, it will be used based on its contribution to solution diversification and quality, ensuring a stable and improvable characteristic of the BOA. Due to the endogenous complexity of the problem, managers need to set planning horizons appropriately based on factors such as the number of production lines, the number of AGVs, and the frequency of material requirements, in order to achieve a balance between planning stability and feasibility.

Future research can be pursued in several directions. Firstly, the incorporation of uncertainty in the production system, such as cycle time and delivery time, could be an interesting direction. Secondly, the consideration of different start times for each flowline within a multi-line production setting could enhance the model's applicability. Although this paper has concentrated on order sequencing across parallel lines, which contributes to reduced total order delays, highly overlapping production schedules across multiple flowlines may pose formidable challenges in material delivery scheduling.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The authors confirm that the data supporting the findings of this study are available in the article and the supplemental file.

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Appendices

Appendix 1. The production scheduling model without considering material delivery

The production model formulated to calculate the U_{min} in section 3.2 is with the objective of minimising the total order delay, see (A1).

$$\max \sum_{j \in J} \max(T_j - d_j, 0) \quad (A1)$$

Its constraints are the same as constraints (2)–(8) in section 3.2, describing the assignment and sequencing of orders. For the temporal relation of order, we set constraint (A2), which means if order j' is assigned to process immediately after order j , then its processing start time can not be earlier than the time when both the process of order j and the mould switch are completed.

$$T_{j'} - p_{j'}^l Q_{j'} \geq T_j + e_{jj'} - \Omega(1 - x_{jj'}), \forall j \in J \cup \{0\} \quad (A2)$$

The two sequential scheduling methods used in section 5.2, both SSG and SSGA, make the production schedule by solving the above production scheduling model. Given the production schedule made by solving the above model, SSG makes the material delivery schedule by solving an optimisation model, too. The material delivery schedule model consists of the objective (1) and constraints (9)–(21) in section 3.2.

Appendix 2. Linearisation of the MIP model

The following is the linearisation of the MIP model proposed in section 3.2.

With auxiliary variable $u1_j$, the objective can be linearised as

$$\min \theta_1 \frac{\sum_{j \in J} u1_j - U_{min}}{U_{max} - U_{min}} + \theta_2 \frac{h - 1}{H - 1} \quad (A3)$$

$$u1_j \geq T_j - d_j, \forall j \in J \quad (A4)$$

With auxiliary variable $u2_{jj'l}$, constraints (8) can be linearised as

$$x_{jj'} \leq \sum_{l \in L} u2_{jj'l}, \forall j, j' \in J \quad (A5)$$

$$u2_{jj'l} \leq z_{jl}, \forall j, j' \in J, l \in L \quad (A6)$$

$$u2_{jj'l} \leq z_{j'l}, \forall j, j' \in J, l \in L \quad (A7)$$

$$u2_{jj'l} \geq z_{jl} + z_{j'l} - 1, \forall j, j' \in J, l \in L \quad (A8)$$

With auxiliary variable $u3_{l'ij'k}^{lijk}$, constraints (16) can be linearised as

$$\begin{aligned} t_{l'ij'k}^d - \tau_{l'ij'k}^o - \sum_{l \in L} \sum_{j \in J} \sum_{i \in I_j^l} \sum_{k \in K_{ij}^l} \tau_{l'ij'k}^i \cdot \gamma_{l'ij'k}^{lijk} \\ - \sum_{l \in L} \sum_{j \in J} \sum_{i \in I_j^l} \sum_{k \in K_{ij}^l} u3_{l'ij'k}^{lijk} \geq 0, \forall l' \in L, \\ j' \in J, i' \in I_{j'}^{l'}, k' \in K_{ij'}^{l'} \end{aligned} \quad (A9)$$

$$u3_{l'ij'k}^{lijk} \leq \Omega \cdot \gamma_{l'ij'k}^{lijk},$$

$$\forall l, l' \in L, j, j' \in J, i, i' \in I_j^l, k \in K_{ij}^l, k' \in K_{ij'}^{l'} \quad (A10)$$

$$u3_{l'ij'k}^{lijk} \leq t_{l'ij'k}^d, \forall l, l' \in L, j, j' \in J, i, i' \in I_j^l, k \in K_{ij}^l, k' \in K_{ij'}^{l'} \quad (A11)$$

$$\begin{aligned} u3_{l'ij'k}^{lijk} \geq t_{l'ij'k}^d - \Omega(1 - \gamma_{l'ij'k}^{lijk}), \forall l, l' \in L, j, j' \in J, i, i' \in I_j^l, \\ k \in K_{ij}^l, k' \in K_{ij'}^{l'} \end{aligned} \quad (A12)$$

With auxiliary variable $u4_{j'l}$, $u5_{jj'}$, constraints (17) can be linearised as

$$t_{l'1j'l}^f - p_{j'l}^l q_{l'1j'l} - \Omega(z_{j'l} - 1) - u4_{j'l} \geq 0, \forall l' \in L, j' \in J \quad (A13)$$

$$u4_{j'l} \geq t_{l'1j'l}^d, \forall l' \in L, j' \in J \quad (A14)$$

$$u4_{j'l} \geq \sum_{j \in J \cup \{0\}} e_{jj'} x_{jj'} + \sum_{j \in J} u5_{jj'}, \forall l' \in L, j' \in J \quad (A15)$$

$$u5_{jj'} \leq \Omega x_{jj'}, \forall j, j' \in J \quad (A16)$$

$$u5_{jj'} \leq t_{l'M''jK_{M''j}^{l'}}^f, \forall j, j' \in J \quad (A17)$$

$$u5_{jj'} \geq t_{l'M''jK_{M''j}^{l'}}^f - \Omega(1 - x_{jj'}), \forall j, j' \in J \quad (A18)$$

With auxiliary variable $u6_{lij}$, constraints (18) can be linearised as

$$t_{lij1}^f - p_j^l q_{lij1} - \Omega(z_{jl} - 1) - u6_{lij} \geq 0, \forall l \in L, j \in J, i \in I_j^l \setminus \{1\} \quad (A19)$$

$$u6_{lij} \geq t_{l,i-1,j,K_{i-1,j}^l}^f - p_j^l Q_j + p_j^l, \forall l \in L, j \in J, i \in I_j^l \setminus \{1\} \quad (A20)$$

$$u6_{lij} \geq t_{lij1}^d, \forall l \in L, j \in J, i \in I_j^l \setminus \{1\} \quad (A21)$$

With auxiliary variable $u7_{lij}$, constraints (19) can be linearised as

$$\begin{aligned} t_{lij}^f - p_j^l q_{lij} - \Omega(z_{jl} - 1) - u7_{lij} \\ \geq 0, \forall l \in L, j \in J, i \in I_j^l, k \in K_{ij}^l \setminus \{1\} \end{aligned} \quad (A22)$$

$$u7_{lij} \geq t_{l,i,j,k-1}^f, \forall l \in L, j \in J, i \in I_j^l, k \in K_{ij}^l \setminus \{1\} \quad (A23)$$

$$u7_{lij} \geq t_{lij}^d, \forall l \in L, j \in J, i \in I_j^l, k \in K_{ij}^l \setminus \{1\} \quad (A24)$$

The domains of the introduced auxiliary variables are as follows.

$$u1, u3, u4, u5, u6, u7 \geq 0 \quad (A25)$$

$$u2 \in \{0, 1\} \quad (A26)$$

Appendix 3. An example of the BRKC operator

Figure A1 illustrates an example of the BRKC operator based on the situation with six orders (named A, B, C, D, E, F) processing on two flow lines. Subgraph (a) displays the original production schedule, where orders A and C are allocated on flow line 1, while orders D, E, B, and F are allocated on flow line 2. In subgraph (b), each order is encoded by a real number following the defined rules. To make the crossover more convenient, real numbers generated in (b) are listed in the alphabetical order of A to F in (c). With the crossover of the current solution and a selected solution, a new encoded solution is generated in (d). The three coloured positions represent the operator's inheritance from the current, while the remaining ones without colour reflect the operator's shaking nature. After the parameterised crossover, the orders are reallocated to two sequences by their coding number, which is illustrated in (e). At last, the real number sequences are decoded into corresponding order sequences in (f).

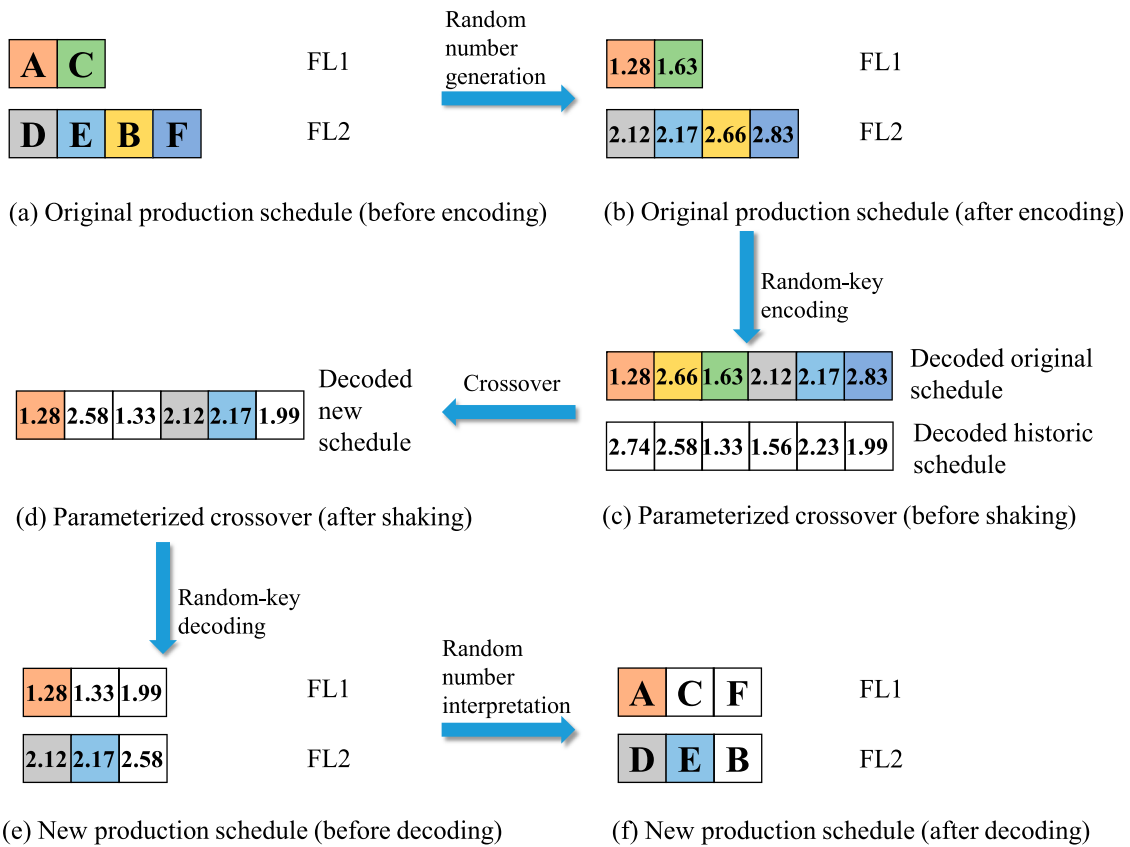


Figure A1. Illustration of the BRKC.